

Radioprotective effects of Zingiber officinale Roscoe (ginger): past, present and future.



Radiation is an important modality in treating people with cancer especially when surgical intervention is impracticable or might debilitate the patient. However, effective use of ionizing radiation is compromised by the side effects that result from radiation-induced damage to normal tissue. The use of radioprotective compounds, which can selectively protect normal tissues against radiation injury is of immense use because in addition to association with protecting the normal tissue, it will also permits use of higher doses of radiation to obtain better cancer control and possible cure. However, till date no ideal radioprotectors are available as most synthetic compounds are toxic at their optimal concentrations. Plants commonly used as dietary and or therapeutic agents have recently been the focus of attention since in most cases they are non-toxic and are easily accepted for human use. Ginger, the rhizomes of *Zingiber officinale* Roscoe (Zingiberaceae), has widely been used as both culinary and medicinal agent. Preclinical studies carried out in the last decade has shown that ginger and its phytochemicals dehydrozingerone, zingerone possess radioprotective effects in laboratory animals and in cultured cells in vitro. The hydroalcoholic extract of ginger rhizome when administered either through intraperitoneal or oral route was effective in protecting against gamma radiation-induced sickness and mortality. The phytochemicals dehydrozingerone and zingerone present in ginger are also shown to protect mice against radiation-induced sickness and mortality. Mechanistic studies have indicated that the free radical scavenging, antioxidant affects, anti-inflammatory and anti-clastogenic effects may contribute towards the observed protection. Additionally, studies with tumor bearing mice have also shown that zingerone selectively protects the normal tissues against the tumoricidal effects of radiation. This review for the first time summarizes the results related to the radioprotective properties and also emphasizes the aspects that warrant future research to establish its activity and utility as a radioprotective agent.